estimated total distance from start node to target node. If this estimate is greater than the value stored for distance travelled in the target mode record continue to rewind. If the estimate is less than this stored value copy the record for the node into the open list as a parent node and continue with the actions listed above for when the target node is not in the open list.

We can consider how the algorithm will progress if the start node is chosen to be town A and the target node to be town D.

With A the parent node, B, C and F become the adjacent nodes whose records are brought into the open list. The calculations are:
For B distance travelled = 13, estimated further distance is 100, total is 113
For C distance travelled = 45, estimated further distance is 51, total 96
For F distance travelled = 20, estimated further distance is 89, total 109
(Note that there are two calculations that have to be made for C and F but in each case the direct route is the shortest.)
B and F are now moved to the closed list and C becomes the new parent. The only adjacent node in the initial list is G which is brought into the open list.
G becomes the parent for the adjacent nodes D and E.
D is identified as the target node. The actual distance of travel from A to D using this route is calculated as 110. When the data has been stored in the record for D this is moved to the initial list.

E is the only adjacent node left in the open list so this automatically becomes the new parent node. The only node in the initial list is D which is again brought into the open list as an adjacent node. The actual distance of travel from A to D is now calculated as 105. This is a lower vale than the one previously calculated so this data is stored in the record for node D. This record is then moved once again into the initial list.



The rewinding will now consider nodes B and F. The calculations for these were carried out earlier. The estimated distances for A to D via B and for A to D via F were found to be 109 and 113 respectively. Both of these values are greater than the actual distance found for a different route so these nodes can be ignored. The problem has been solved. The shortest route has been found to be A-C-G-E-D with a total distance of 105.

**Discussion Point:**
Do you know what a heuristic function is?

**Discussion Point:**
The techniques discussed here can be used for path-finding when there are obstacles. Can you find some information about how this is approached?

## 22.03 Machine learning

The requirements for **machine learning** can be summarised as:

- a computer-based system has a defined task or tasks to perform
- knowledge is acquired through the experience of performing the tasks
- as a result of this experience and the knowledge gained the performance of future tasks is improved.

The ability to learn from experience is an indication of intelligence. Machine learning is therefore one of the many individual approaches defined under the umbrella of artificial intelligence.

There are a number of ways to describe how the learning can take place. Three of these will be considered here.

In **unsupervised learning** the system has to draw its own conclusions from its experience of the results of the tasks it has performed. For this, algorithms are needed that can organise or categorise the knowledge acquired. An example is where 'conceptual clusters' are identified which are based on a hierarchical framework. In this approach the knowledge is initially all placed in the root of a tree structure. Then, depending on attributes of the knowledge, selected groups are moved into branches of the tree.

Nowadays unsupervised learning is a dominant activity. Powerful computer systems having access to massive data banks are regularly used to make decisions based on previous actions recorded. We all have our activity on the world wide web recorded and stored. This stored data is then used to make decisions about what products or services should be recommended to us in future Internet use. There is no theoretical framework for this; it is rather as though the intelligence is built-in to the data.

In **supervised learning** the system is fed knowledge with associated classification. For example, an AI program might be under development for marking exam paper questions. In the supervised learning, answers to examination questions could be provided together with a grade for each one or with categorised comments.

A special case of supervised learning is where an expert system is being developed. An expert system always has a focus on a narrowly defined domain of knowledge. In this case human experts are given samples of data requiring analysis. The experts provide the conclusions to be drawn from the data. The data and conclusions are input to the knowledge base. The effectiveness of the system can be tested by a human expert providing sample data and checking the accuracy of the conclusions provided by the expert system. If performance is poor, then further data and conclusions are input to the system. Although an expert system is an example of AI it is not an example of machine learning because there is no expectation that the system will improve its performance unaided.



**Reinforcement learning** has some features similar to unsupervised learning and other features similar to supervised learning. The method has its own specific vocabulary. The following statements use this vocabulary in describing aspects of how a reinforcement learning algorithm works.

- An agent is learning how best to perform in an environment.
- The environment has many defined states.
- At each step the agent takes an action.
- An agent has a policy that guides its actions.
- The policy is influenced by the recorded history and the knowledge of the current state of the environment.
- An action changes the environment to a new state.
- The agent receives a reward following an action which is a measure of how effective the action was in relation to the achievement of the overall goal.
- The policy will guide the agent in deciding whether the next action should be exploiting knowledge already known or exploring a new avenue.



**KEY TERMS**

**Machine learning:** where a system improves its performance through analysis of previous performance

**Unsupervised learning:** where the machine learning takes place entirely through the system analysing and categorising the available data

**Supervised learning:** where sample data is supplied to the system with associated data relating to the outcome of its use

**Reinforcement learning:** where an agent learns by receiving graded rewards for actions taken

In summary, the aim is to maximise the reward values by improving the quality of the policy. It is a trial-and-error search for optimum performance. It requires many repeated attempts at the same problem.

This description is an abstraction. The concept becomes clearer if some instances of the application of the approach are considered. One area of application is playing logic games such as backgammon. Another is robotics where a robot has to learn how to become effective at a task. Another option is where the machine has to learn how to navigate a maze. In this last case when the agent chooses a left turn which will eventually lead to the destination the reward is given a positive numeric value. If instead it chooses a right turn the reward is given a negative value.

## Regression analysis methods

In some applications the aim of the AI is to predict and provide, as output numerical values for some defined quantity, on the basis of data values for different quantities that have been input to the AI algorithm. If **regression analysis** is to be used, the first step is for the system to be provided with some actual values for both the input data and for what will become output data when the AI system is operational. This data can be used to investigate if there is any correlation between these sets of values. If a correlation is established which can be represented by a mathematical formula, then this formula can be used to output predicted values when new data is input.

KEY TERM

**Regression analysis:** finding a mathematical function that provides the best fit to the actual outcomes when outcomes are calculated from previous inputs



The simplest application of regression analysis is when values for only one quantity are to be input and when a linear relationship is expected between these values and the values to be predicted. An example could be an AI system being used to predict marks for candidates in an A Level Computer Science exam. There may be an expectation that there would be a correlation between a candidate's marks in an A level Computer Science exam and their marks in an IGCSE Mathematics exam. Figure 22.04 shows what might be found when some historic data is input and analysed.

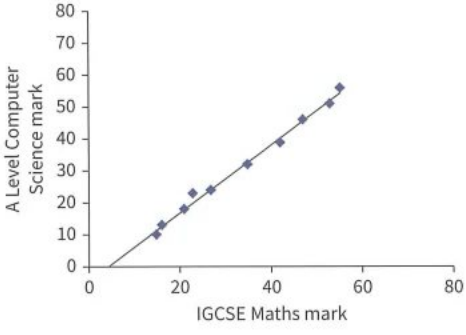



Figure 22.04 An example of a linear regression analysis

There is good correlation between the two sets of marks. The straight line in Figure 22.04 is the one that has been calculated as the best fit to the data. The formula for the line can sensibly be used to predict future marks for the A level Computer Science paper from the marks scored in an IGCSE Mathematics paper.

The regression analysis could be more complicated. For example, a similar fitting to a mathematical formula can be carried out if marks for three different exams are used as input. In other cases, a non-linear relationship might be appropriate as might happen when a prediction of future sales of a new product was needed where the growth in sales was expected to be exponential.

## 22.04 Artificial neural networks

The neural networks in our brains provide our intelligence. It therefore seems obvious that artificial neural networks should be considered as a foundation for artificial intelligence systems. Figure 22.05 shows a representation of two nerve cells.

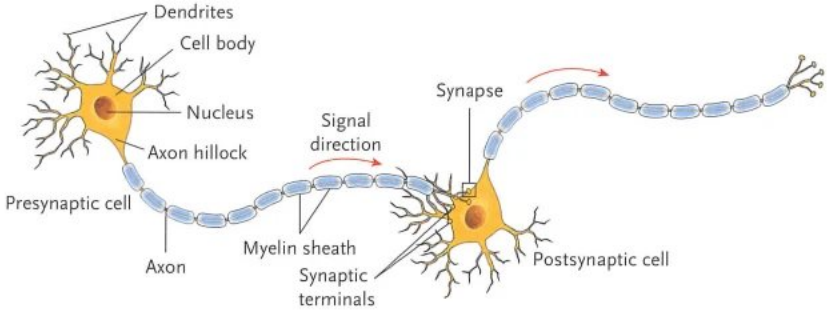





Figure 22.05 Two nerve cells showing how a signal is transmitted

At the one end of a nerve cell there are many dendrites which can receive signals. At the other end of the cell there are many axon terminal buttons that can transmit signals. The synapse is the region between an axon terminal button and a dendrite which contains neurotransmitters. When a nerve cell receives input signals the voltage in the axon increases. At some threshold value of this voltage neurotransmitters are activated and signals are sent to the dendrites of adjacent cells.

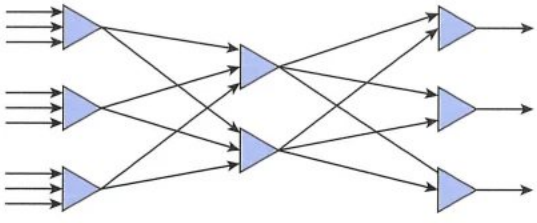


Figure 22.06 A schematic representation of a simple artificial neural network

An artificial neural network could be created in software or hardware. The components of the network can be represented by a diagram as illustrated in Figure 22.06. The triangles are the nodes in the network which represent artificial neurons. (Sometimes these are represented as circles). In general, a node can receive one or more inputs and can provide an output to one or more of the other nodes. The modelling of the action of the node involves applying a weighting factor to each input. The weighted input values are summed and then an activation function is used to compute a value for the output of the node. If the input is not a numerical value it must be converted to one so that the weighted values can be summed.

Figure 22.06 shows a very simple network structure consisting of three layers. The column of three nodes on the left receive input. The column on the right provides output. The two nodes in between form what is referred to as a hidden layer. Some artificial neural networks will contain several hidden layers.

An example of an AI system using an artificial neural network is one which estimates the cost-effectiveness of batteries based on the initial price paid and the lifetime of use. Each input node would represent one example of a battery. The inputs would be specific data that identified the battery and the price. One of the nodes in the hidden layer could be concerned with the type of device the battery was to be used in. The other node in the hidden layer could relate to the type of user of the device. Each node in the output layer would compute an estimated value for the cost per unit of time for a specific battery.

In this system there are adjustable factors for each node. These are the weighting factors for each input and the activation function. When the initial learning is taking place, these adjustable factors need to be tuned to achieve the optimum predictive capability of the system. The method that can be used for this is **back propagation of errors**. There is a need for some battery lifetime data from some real use of the batteries. The AI system is created with a set of trial values for all of the adjustable factors. The system is run with the input data that matches the real use. There will then be an error identified for each output node. The error is the difference between the output value and the real-use value for the battery lifetime. The system is then re-run with different values for the weighting factors and activation functions applying to the output nodes. This will determine the dependency of the accuracy of the output value on the adjustable factors associated with the performance of the nodes in the output layer.



KEY TERM

**Back propagation of errors:** An algorithm for machine learning that optimises the values for parameters which are adjustable. It is applied first to the nodes in the output layer and then works backward through the nodes in hidden layers until finally the input nodes are considered.

The next step is to carry out a similar investigation for the nodes in the hidden layer. Finally, the adjustable factors for the nodes in the input layer are tackled. This process of learning should now be sufficient for the system to be applied to some new input data to predict the cost-effectiveness of some different batteries. When more real-use data becomes available then the back propagation of errors learning can be applied again to achieve improved performance of the system.

### Extension Question 22.01

An early successful application of an artificial neural network was NETtalk. This was able to take a text as input and output a synthesised sound reading of the text. The network had 7 times 29 input nodes and 26 output nodes. Can you investigate why that number of input nodes was chosen and what was the function of some of the 80 nodes in the hidden layer?

## Deep Learning

It is understood that in the brain there is a layer structure of neurons where lower layers have readily understandable functions but where higher layers are concerned with more abstract processing of information. With the increasing computing power now available, artificial neural networks are being introduced with large numbers of hidden layers which are attempting to achieve something similar. These are known as **Deep Learning** systems.

KEY TERM

**Deep learning:** where a system uses an artificial neural network with an exceptionally large number of hidden layers

**Reflection Point:**

Could you create a hierarchical chart to show how the various approaches discussed in this chapter are related to each other?

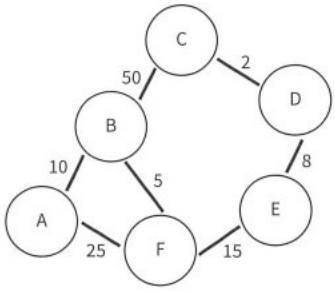
## Summary

- A graph can be constructed from nodes and edges where the edges carry numerical value labels.
- Algorithms are available to find the shortest path between two nodes in a graph.
- Machine learning can be supervised or unsupervised.
- Regression analysis involves finding a mathematical equation which is a best fit to sample data.
- Artificial neural networks are modelled using nodes which receive input and provide output.
- Back propagation of errors can be used for machine learning using artificial neural networks.



## Exam-style Questions

**1** The diagram below shows a graph representing the cost of journeys between railway stations identified by A, B, C, D, E and F.




Dijkstra's algorithm is to be used to find the total cost for journeys from station A to each of the other stations. A record structure is to be used to store for each station the cost for the travel so far and the list of stations so far visited in the order visited. Complete the table below to record the progress of the algorithm by identifying which nodes are in the ShortestPath set and what would be stored in the record at each step of the algorithm.